Akhil Vanukuri

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Research Interests

Cryptography, Security.

Education

PhD in Computer Science	University of Wisconsin, Madison CGPA: 3.893 / 4.0	Aug, 2024 - *present*
MS (By Research) in C.S.E[1]	Indian Institute of Technology, Madras Advisor: Shweta Agrawal Thesis: Cryptographic applications of k-SUM CGPA: 8.80 / 10.0	Jan, 2023 - July, 2024
MSc. in Computer Science	Chennai Mathematical Institute Thesis advisor: Shweta Agrawal Thesis: Fully Homomorphic Signatures CGPA: 8.81 / 10.0	Dec, 2020 - Jul, 2022
B.Tech. in C.S.E	Manipal Institute of Technology, Manipal CGPA: 8.22 / 10.0	Jul, 2016 - Aug, 2020

Conference Publications

k-SUM in the Sparse Regime
Shweta Agrawal, Sagnik Saha, Nikolaj I. Schwartzbach, Akhil Vanukuri, Prashant Nalini Vasudevan
Accepted at CRYPTO'24: The 44^{th} Annual International Cryptology Conference

Experience

• **Project Associate**	Indian Institute of Technology, Madras	May 2022 - May 2024
• **Research Intern**	National University of Singapore, Singapore Internship Supervisor: Prashant Nalini Vasudevan	Jun 2023 - Aug 2023
• **Student Trainee**	Samsung Research and Development Institute, Bangalore	Jan 2020 - Jun 2020
• **Research Intern**	R. C. Bose Centre for Cryptology, and Security, ISI Kolkata Internship Supervisor: Sushmita Ruj	May 2019 - Jul 2019

Professional Activities

Sub-reviewer for Crypto 2023, PKC 2024, Crypto 2025.

Research Projects

- **Study of k-SUM problem in the sparse regime** — IIT Madras and NUS, Singapore
  Studied properties of the k-SUM problem in the sparse regime and constructed cryptographic schemes using it

- **Cloud Data Auditing Using Proofs of Retrievability and Replication** — ISI, Kolkata
  Implemented proof of concept (using gmp, pbc library in C) and made contributions w.r.t mathematical aspects of a public verifiable privacy-preserving proof of replication scheme
  (Co-author of a manuscript with Nishant Dhanaji Nikam and Sushmita Ruj).

[1] Computer Science and Engineering

- **Dehazing** Samsung R & D, Bangalore
  Designed and implemented a novel CNN-based solution for dehazing of a hazy image (using PyTorch).

Scholarships

- 2020-2022 Cognizant Foundation Scholarship

Talks

- **k-SUM in the sparse regime** Bangalore Crypto Day, 2024

Relevant Courses[2]

Structure vs Hardness in Cryptography
Grade : 10/10

Foundations of Cryptography
Grade : 9/10

Topics in Cryptography
Grade : 9/10

Modern Complexity Theory
Grade : 8/10

Introduction to Information Security
Grade : A

Linear Algebra (Accelerated Honors)
(Audited)

Computability and Complexity
Grade : 9/10

Quantum Algorithms and Cryptography[3]
Grade : 10/10

Algebra II (Abstract Algebra)
Grade : 8/10

User-Centric Security & Privacy
Grade : AB

Computer Networks
Grade : 9/10

Structural Graph Theory
Grade : 8/10

Teaching Assistance

- **CS1100: Introduction to Programming**
  Lead Teaching Assistant for both theory and lab component of the course at IIT, Madras.
  Awarded Star TA award for the work done.
- **CS577: Introduction to Algorithms**
  Teaching Assistant at UW-Madison
- **CS435: Introduction to Cryptography**
  Teaching Assistant at UW-Madison
- **CS240: Discrete Mathematics**
  Teaching Assistant at UW-Madison

Other Projects [4]

- **Parallelization of AES encryption**
  Implemented a parallelized version of AES using CUDA, OpenGL.
- **Placement Portal Website**
  Developed a placement management website using C# and MySQL.
- **Parser for Lua**
  Developed a parser for the programming language Lua using C, Flex, Bison.

[2] Click on the subjects to see the topics covered
[3] under the name "Quantum Algorithms and Complexity" in the transcript
[4] These projects are part of undergrad course-requirement and have been done in collaboration with other classmates.

- **Movie Database App**
  Developed an application that allows information retrieval from a database storing film-related information using Java, SQL.

Miscellaneous

- Programming Languages: C, JAVA, Python, Haskell, SQL
- Other relevant technical skills: CUDA, OpenGL, MPI, LaTeX, HTML, Git.
- Libraries: GMP (GNU Multiple Precision Arithmetic), PBC (Pairing Based Crypto), PyTorch